

Abolfazl Abbasi

Personal info

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Education

Associate of Software Engineering
Technical Vocational College of
Qom
2022 - 2024

B.S. Software Engineering
Technical Vocational College of
Qom
2024 – Present (Studying)

Technical Skills

Languages & Libraries:

Python (asyncio, Pandas, NumPy,
scikit-learn, Transformers)

AI/ML & NLP:

LLMs (HuggingFace Transformers)
Text Classification
Semantic Search
Prompt Engineering
Model Evaluation & Fine-tuning

Data Engineering:

MongoDB, Redis, PostgreSQL
Web Scraping (Selenium,
BeautifulSoup)
Data Pipelines, Preprocessing

DevOps & Tools:

Git/GitHub, Docker, Linux, vim

AI Engineer | NLP & LLM Systems | Financial Bot

Professional Summary

AI Engineer with hands-on experience in NLP, LLMs, and algorithmic trading. Specialized in building scalable systems for legal, financial, and e-commerce domains using Python and Transformers. Skilled in prompt engineering, model fine-tuning, semantic search, and automation, PostgreSQL, and Docker.

Work Experience

- **AI Engineer & Python Developer | Mahroyan Software Industry Co**
July 2023 - April 2025
 - Redesigned a scraping system into a modular, scalable architecture for collecting social media data across X, Facebook, and Instagram.
 - Fine-tuned a BERT model for emotion classification using ChatGPT-generated samples and analyst feedback to improve accuracy across 8 emotions and 3 polarities.
 - Built a Persian legal assistant using RAG, vector search, and prompt engineering with few-/zero-shot learning for case analysis and contract generation. deployed via Streamlit and Gradio.
- **AI & Trading Systems Developer(quant) | FintechPlus (Freelance)**
 - Built a labeled financial chart dataset and fine-tuned a YOLOv8 model for detecting and designed a real-time pipeline for trading integration, achieving 97% detection accuracy.
 - Developed a real-time automated Forex trading bot with coded risk management strategies and decision logic. implemented news-based trading suspension using Redis for caching and PostgreSQL for long-term data storage.
- **AI & Automation Developer | MaralBranding (Freelance)**
 - Built a modular e-commerce chatbot using RAG and LLMs to answer customer queries from product data. integrated semantic search
 - Built a Windows desktop app with PyQt5 for scraping and managing business data from Balad.ir; enabled search, filtering, and export for non-technical users.

Projects

Social Botnet Intelligence System

- Designed and deployed a **distributed botnet** across major social media platforms including X (Twitter), Facebook, and Instagram
 - Each bot simulated a **realistic virtual persona** with unique behavioral patterns for commenting, liking, sharing, and replying to posts
 - Implemented an module for collecting posts, comments, hashtags, and trending content using topic and language filters
 - Built a centralized control system to coordinate bot activities base of targeted groups
- Technologies: Python, Selenium, Asyncio, Flask, MariaDB, BeautifulSoup

Sentiment classification & Emotion Detection System

- Fine-tuned a BERT-based model on labeled financial data for multi-label emotion classification (positive, negative, neutral + 8 emotional states).
- Addressed class imbalance using weighted loss and ChatGPT-generated negative samples to augment underrepresented emotions.
- Established a human-in-the-loop cycle where senior analysts reviewed and corrected model outputs, enabling continuous refinement and retraining for improved accuracy.

Technologies: HuggingFace Transformers, ChatGPT API, prompt engineering

Legal Reasoning & Contract Chatbot System

- Designed a RAG-based legal QA system using a custom Persian dataset built from scraped Q&A legal forums and official documents.
- Integrated semantic search with vector databases (ChromaDB) to retrieve relevant legal texts and precedents for case-based question answering.
- Developed legal advisory and contract generation modules using prompt engineering combined with few-shot and zero-shot learning techniques for personalized output.
- Built an interactive moot court simulation to emulate legal argumentation and support training and exploration.
- Deployed production-ready web interfaces using Streamlit and Gradio, enabling fast interaction and deployment.

Technologies: OpenAI API, HuggingFace Transformers, Sentence Transformers, ChromaDB, BeautifulSoup, LangChain, Streamlit, Gradio, Docker

Chart Pattern Detector using Object Detection

- Built a computer vision pipeline to detect technical chart patterns in candlestick chart images
- Fine-tuned pre-trained YOLOv8 object detection models on a custom-labeled dataset of financial chart patterns
- Created custom annotation workflows and preprocessing routines for high-precision detection
- Integrated extracted pattern data into automated trading strategy logic as structured visual features
- Technologies: YOLOv8, OpenCV, Python, Matplotlib

Automated Forex Trading System

- Engineered a bot mechanical trading system using MetaTrader EA (via Python), focused on price action strategies including RSI divergence and candlestick formations and SMC
- Integrated automated risk management: session filters, adaptive lot sizing, trailing stops
- Implemented real-time news monitoring module to detect high-impact economic events and automatically pause trading activity to avoid volatile conditions
- Utilized Redis as a low-latency cache for recent tick data and news events; stored long-term trading logs and market snapshots in PostgreSQL
- System designed for hands-off execution with clear entry/exit logic and rule-based decision-making
- Technologies: Python, MetaTrader, Redis, PostgreSQL, Forex Factory, Asyncio

Custom E-Commerce Chatbot System

- Built a ChatGPT-powered assistant for customer support by integrating product data and FAQs to deliver accurate, context-aware responses.
- Implemented a RAG-based retrieval system combined with semantic search to match user questions with product answers.
- frequent question detection to improve response consistency and reduce redundancy.

Technologies: Python, OpenAI API (ChatGPT), LangChain, Sentence Transformers, Flask, PostgreSQL, Docker

Data Collection & Desktop Tool Developer | Balad Business Scraper

- Developed a web scraping system to extract structured business information (name, category, address, phone number, etc.) from Balad.ir for use in commercial data aggregation.
- Built a desktop application with an intuitive UI to search, filter, and export business data. designed for non-technical users and compatible with Windows OS.

Technologies: Python, Requests/BeautifulSoup, PyQt5, Pandas, Windows EXE packaging.